

Construction of Numerical Energy Functions for Lossy Transient Stability Models

7.1 INTRODUCTION

The existing numerical energy functions for lossy power system stability models are not analytical in the sense that they are not well-defined functions. They all contain two major terms: the analytical terms (i.e., path-independent terms) and the path-dependent terms. The path-dependent terms are not well-defined functions and require numerical approximations to become well-defined. The so-called ray approximation scheme and the trapezoidal approximation scheme are popular schemes for numerically approximating the path-dependent terms (Fouad and Vittal, 1991; Pai, 1989; Pavella and Murthy, 1994; Qiang and Zhong, 2005; Sasaki, 1979).

In this chapter, methodologies for constructing numerical energy functions for lossy power system stability models will be presented. At present, only two methods are available. One is based on the so-called first integral principle, while the other is based on a two-step procedure. Numerical energy functions derived by these two methods all contain path-dependent terms which need to be evaluated by numerical approximation schemes.

It will be shown that the so-called ray approximation scheme leads to a numerical ill-conditioned problem during the numerical approximation of path-dependent terms. A scheme to eliminate this numerical ill-conditioning will be presented. However, even after this elimination, the ray approximation scheme can give unacceptable results in the critical clearing time (CCT) estimate. It is recommended that the trapezoidal rule be used to evaluate the path-dependent terms of the numerical energy function. Numerical studies of these two approximation schemes will be conducted and the numerical results will be compared.

Finally, improved numerical approximation schemes for evaluating the path-dependent terms will be presented. These improved numerical approximation schemes are based on multipoint integrations. It will be numerically shown that the improved schemes are superior to the two popular approximation schemes: ray approximation and trapezoidal approximation schemes.

7.2 A TWO-STEP PROCEDURE

The existing (numerical) energy functions for lossy power system stability models are not analytical in the sense that they are not well-defined functions. Numerical energy functions were introduced to resolve the difficulty of constructing (analytical) energy functions via, for example, the inclusion of the first integral of the vector field introduced by the transfer conductances.

A general two-step procedure to derive a numerical energy function for a lossy transient stability model is described as follows:

Step 1. Construct an analytical energy function for the underlying model without the lossy terms, such as transfer conductance.

Step 2. Construct a numerical energy function by adding the first integral of the vector field introduced by the transfer conductances to the analytical energy function constructed in Step 1.

The above two-step procedure leads to a numerical energy function, which has two terms: an analytic energy function and a path-dependent energy function related to lossy terms.

We next give an illustration of the above two-step procedure. Many existing lossy network-reduction transient stability models can be written in the following compact form (Chu and Chiang, 1999):

$$\begin{aligned} T\dot{x} &= -\frac{\partial U}{\partial x}(x, y) + g_1(x, y) \\ \dot{y} &= z \\ M\dot{z} &= -Dz - \frac{\partial U}{\partial y}(x, y) + g_2(x, y), \end{aligned} \tag{7.1}$$

where $x \in R^n$, y , and $z \in R^m$, T , M , and D are positive diagonal matrices; $g_1(x, y)$ and $g_2(x, y)$ represent the transfer conductance of the reduced network. $U(x, y)$ is a smooth function. We apply Step 1 of the procedure to derive an analytical energy function; that is, $W(x, y, z) = K(z) + U(x, y) = \frac{1}{2}z^T Mz + U(x, y)$ for the following lossless model:

$$T\dot{x} = -\frac{\partial U}{\partial x}(x, y)$$

$$\dot{y} = z$$

$$M\dot{z} = -Dz - \frac{\partial U}{\partial y}(x, y).$$

By adding the first integral of the vector field introduced by the transfer conductances to the analytical energy function constructed in Step 1, it leads to a numerical energy function that has two terms: an analytical energy function, $W_{ana}(x, y, z) = \frac{1}{2} z^T M z + U(x, y)$, and a path-dependent potential energy, $U_{path}(x, y, z)$:

$$\begin{aligned} W_{num}(x, y, z) &= W_{ana}(x, y, z) + U_{path}(x, y) \\ &= K(z) + U(x, y) + U_{path}(x, y). \end{aligned} \quad (7.2)$$

The above numerical energy function contains path-dependent terms, which can be approximately evaluated using two trajectory approximation schemes: (1) ray approximation and (2) trapezoidal approximation.

We next explain how to evaluate path-dependent terms using ray approximation. For example, we evaluate the following path-dependent term:

$$U_1 = \sum_{i=1}^n \int G_{ii} V_i^2 d\theta_i, \quad (7.3)$$

in which the state variables involved in the integration can be approximated by the following ray:

$$\begin{aligned} V_i &= V_{io} + \lambda \Delta V_i \\ \theta_i &= \theta_{io} + \lambda \Delta \theta_i. \end{aligned} \quad (7.4)$$

By substituting Equation 7.3 into Equation 7.4, it follows that

$$\begin{aligned} U_1 &\cong \sum_{i=1}^n \int_0^1 G_{ii} (V_{io} + \lambda \Delta V_i)^2 \Delta \theta_i d\lambda \\ &= \frac{G_{ii} \Delta \theta_i}{3} (V_i^2 + V_i V_{io} + V_{io}^2). \end{aligned} \quad (7.5)$$

For another example, we evaluate the following path-dependent term:

$$U_2 = \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n G_{ij} \int V_i V_j \cos \theta_{ij} d\theta_i. \quad (7.6)$$

By using the ray approximation, U_2 can be rewritten as follows:

$$\begin{aligned}
U_2 &= \sum_{i=1}^n \sum_{j=i+1}^n G_{ij} \int V_i V_j \cos \theta_{ij} d(\theta_i + \theta_j) \\
&\cong \sum_{i=1}^n \sum_{j=i+1}^n G_{ij} (\Delta \theta_i + \Delta \theta_j) \left[\begin{aligned} &\frac{V_{io} V_{jo}}{\Delta \theta_{ij}} (\sin \theta_{ij} - \sin \theta_{ijo}) \\ &+ \frac{V_{io} \Delta V_j + V_{jo} \Delta V_i}{\Delta \theta_{ij}^2} (\cos \theta_{ij} - \cos \theta_{ijo} + \Delta \theta_{ij} \sin \theta_{ij}) \\ &+ \frac{\Delta V_i \Delta V_j}{\Delta \theta_{ij}^3} (2 \Delta \theta_{ij} \cos \theta_{ij} + (\Delta \theta_{ij}^2 - 2) \sin \theta_{ij} + 2 \sin \theta_{ijo}) \end{aligned} \right] \text{ and}
\end{aligned} \tag{7.7}$$

$$\begin{aligned}
U_3 &= \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n G_{ij} \int V_j \sin \theta_{ij} dV_i \\
&\cong \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n G_{ij} \Delta V_i \left[\begin{aligned} &\frac{V_{jo}}{\Delta \theta_{ij}} (\cos \theta_{ijo} - \cos \theta_{ij}) \\ &+ \frac{\Delta V_j}{\Delta \theta_{ij}^2} (\sin \theta_{ij} - \sin \theta_{ijo} - \Delta \theta_{ij} \cos \theta_{ij}) \end{aligned} \right].
\end{aligned} \tag{7.8}$$

Another scheme to approximate the path-dependent terms is based on the trapezoidal rule, which is a special case of the closed Newton–Cotes formulae for numerical integration. This rule approximates the integration via the following equation:

$$\int_a^b f(x) dx \cong \frac{b-a}{2} (f(a) + f(b)):$$

Applying the trapezoidal rule to the following term, it follows that the term at the k th time step becomes

$$\begin{aligned}
U_4(k) &= \sum_{i=1}^n \int G_{ii} V_i^2 d\theta_i \\
&\cong \sum_{i=1}^n \frac{G_{ii}}{2} (V_i(k)^2 + V_i(k-1)^2) (\theta_i(k) - \theta_i(k-1)) + U_4(k-1),
\end{aligned} \tag{7.9}$$

where $U_4(0) = 0$.

Applying the trapezoidal rule to the following term, it follows that

$$\begin{aligned}
U_5(k) &= \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n G_{ij} \int V_i V_j \cos \theta_{ij} d\theta_i \\
&\cong \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{G_{ij}}{2} (V_i(k) V_j(k) \cos \theta_{ij}(k) + V_i(k-1) V_j(k-1) \cos \theta_{ij}(k-1)) \\
&\quad \times (\theta_i(k) - \theta_i(k-1)) + U_5(k-1),
\end{aligned} \tag{7.10}$$

where $U_5(0) = 0$.

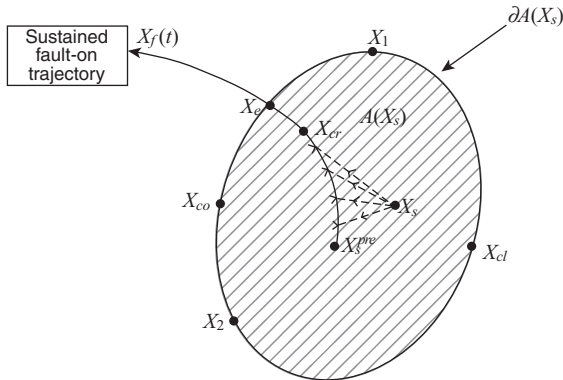


Figure 7.1 Illustration of calculating numerical energy function along the fault-on trajectory using ray approximation.

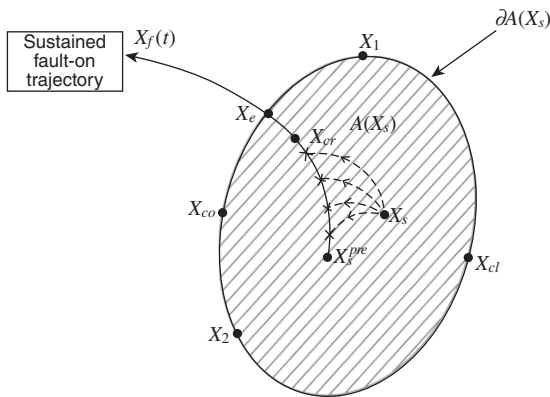


Figure 7.2 Illustration of calculating numerical energy function along the fault-on trajectory using trapezoidal approximation.

Hence, the numerical energy function contains an analytical energy function, which is a path-independent term, and a nonanalytical function, which depends on the integration path. The path-dependent term can be approximated by applying the trapezoidal rule or by the ray approximation scheme. The differences between these two approximations can be small or noticeable depending on the nature of the fault-on trajectory. An illustration of these two approximation schemes in evaluating path-dependent terms is shown in Figures 7.1 and 7.2, respectively.

7.3 FIRST INTEGRAL-BASED PROCEDURE

The purpose of this section is to present a procedure based on the first integral principle to derive an energy function for a general lossy transient stability model. In order to build the complete transient stability model in a systematic manner, the bus number sequence and relevant interfacing variables are defined as follows:

- **Network Bus.** Generator buses are numbered 1, 2, ..., n . Load buses are numbered $n + 1$, $n + 2$, ..., $n + m$.

- **Bus Voltages.** The voltage magnitudes and phase angles are represented as $V_1, V_2, \dots, V_{n+m}$ and $\theta_1, \theta_2, \dots, \theta_{n+m}$, respectively.
- **Power Generations.** The generations from the generator terminal buses into the network are represented by $P_{Gi} + jQ_{Gi}$ for $i = 1, \dots, n$. If generator resistance is neglected, then P_{Gi} is equal to the electrical power P_{ei} . And $P_{Gi} = Q_{Gi} = 0$ for $i = n + 1, \dots, n + m$.
- **Static VAR Compensation.** The reactive power supplied by the static VAR compensator installed at bus i is represented as $Q_{SVCi} = B_{SVCi} V_i^2$, where B_{SVCi} is the equivalent shunt susceptance, and $B_{SVCi} = 0$ if there is no SVC device at bus i .
- **Loads.** Bus loads are represented by $P_{Li} + jQ_{Li}$ for $i = 1, \dots, n + m$. Both P_{Li} and Q_{Li} are voltage-dependent ZIP models; that is,

$$P_{Li} = P_{Li}^{(0)} \left[\alpha_{Pi} + \beta_{Pi} \left(\frac{V_i}{V_i^{(0)}} \right) + \gamma_{Pi} \left(\frac{V_i}{V_i^{(0)}} \right)^2 \right] \quad \alpha_{Pi} + \beta_{Pi} + \gamma_{Pi} = 1$$

$$Q_{Li} = Q_{Li}^{(0)} \left[\alpha_{Qi} + \beta_{Qi} \left(\frac{V_i}{V_i^{(0)}} \right) + \gamma_{Qi} \left(\frac{V_i}{V_i^{(0)}} \right)^2 \right] \quad \alpha_{Qi} + \beta_{Qi} + \gamma_{Qi} = 1$$

for $i = 1, \dots, n + m$.

- **Admittance Matrix.** The original network admittance matrix is denoted as $Y = G + jB$, where $G, B \in R^{(n+m) \times (n+m)}$, or $Y_{ij} = G_{ij} + jB_{ij}$, $i = 1, \dots, n + m, j = 1, \dots, n + m$.

It should be noted that the generator internal reactances are not included in the above network admittance matrix.

- **Network Equation.** The network equation is represented by the power flow equation:

$$\sum_{j=1}^{n+m} V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) + P_{Li} - P_{Gi} = 0$$

$$\sum_{j=1}^{n+m} V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) + Q_{Li} - Q_{Gi} - Q_{SVCi} = 0.$$

$i = 1, \dots, n + m$. It should be noted that for $i = n + 1, \dots, n + m$, $P_{Gi} = Q_{Gi} = 0$. And $Q_{SVCi} = 0$ if bus i is not installed with an SVC device.

The complete transient stability model is composed of mathematical equations describing the system components such as generators, control systems, and transmission networks and loads. Therefore, the energy function W to be derived can be viewed as a combination of several terms related to each component in the model:

- W_{KE} : the kinetic energy associated with generator rotating inertias and speeds,
- U_{GEN} : the potential energy associated with generator circuit equations,

- U_{NET} : the potential energy associated with network equations and loads,
- U_{AVR} : the potential energy associated with automatic voltage regulation systems,
- U_{PSS} : the potential energy associated with power system stabilizers,
- U_{GOV} : the potential energy associated with generator speed governors, and
- U_{SVC} : the potential energy associated with the static VAR compensators.

In summary, the energy function can be written as

$$W = W_{KE} + U_{GEN} + U_{NET} + U_{AVR} + U_{PSS} + U_{GOV} + U_{SVC}.$$

Because the interaction between generators and loads is closely related to the network equations, we derive W_{KE} , U_{GEN} , and U_{NET} simultaneously and denote their sum as W_{G-N} ; that is,

$$W_{G-N} = W_{KE} + U_{GEN} + U_{NET}.$$

Hence, the total energy function can be rearranged as

$$W = W_{G-N} + U_{AVR} + U_{PSS} + U_{GOV} + U_{SVC}.$$

This arrangement is beneficial because it would produce more analytical terms. Based on the first integral principle, the various parts of this energy function will be derived in the following. We next illustrate a procedure to derive an energy function associated with generators and network equations. This procedure is based on the first integral principle.

7.3.1 Energy Function Associated with Swing Equations

The transient energy function associated with the swing equation is calculated as follows:

$$\begin{aligned} W_s &= \int \sum_{i=1}^n \left(M_i \frac{d\omega_i}{dt} \left(-P_{mi} + \frac{M_i}{M_T} P_{col} \right) + P_{Di} \right) \frac{d\delta}{dt} dt \\ &= \sum_{i=1}^n \frac{1}{2} M_i \omega_i^2 - \sum_{i=1}^n \int P_{mi} d\delta_i + \sum_{i=1}^n \int P_{ei} d\delta_i + \sum_{i=1}^n \int D_i \omega_i d\delta_i. \end{aligned}$$

The first term represents the kinetic energy, and the other three terms represent the potential energy due to mechanical, electrical and damping powers.

7.3.2 Energy Function Associated with Power Flow Equation

Multiplying the real power flow equation by $\frac{d\theta_i}{dt}$ and summing over all the $n + m$ equations, we get

$$\sum_{i=1}^{n+m} \sum_{j=1}^{n+m} V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) \frac{d\theta_i}{dt} + \sum_{i=1}^{n+m} P_{Li}(V_i) \frac{d\theta_i}{dt} - \sum_{i=1}^n P_{Gi} \frac{d\theta_i}{dt} = 0.$$

Multiplying both sides of the reactive power flow equation by $\frac{1}{V_i} \frac{dV_i}{dt}$ and summing over all the $n + m$ equations, we obtain

$$\sum_{i=1}^{n+m} \sum_{j=1}^{n+m} V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \frac{dV_i}{dt} + \sum_{i=1}^{n+m} \frac{Q_{Li}}{V_i} \frac{dV_i}{dt} - \sum_{i=1}^n \frac{Q_{Gi}}{V_i} \frac{dV_i}{dt} = 0.$$

Integrating the above two equations and summing them over them, we get

$$\begin{aligned} W_{PF} = & \int \sum_{i=1}^{n+m} \sum_{j=1}^{n+m} [V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) d\theta_i + V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) dV_i] \\ & + \int \sum_{i=1}^{n+m} \left(P_{Li} d\theta_i + \frac{Q_{Li}}{V_i} dV_i \right) - \int \sum_{i=1}^n \left(P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i \right). \end{aligned}$$

By carrying out this integration, we have the following equation:

$$\begin{aligned} W_{PF} = & -\frac{1}{2} \sum_{i=1}^{n+m} B_{ii} (V_i^2 - V_i^{02}) - \sum_{i=1}^{n+m} \sum_{j=i+1}^{n+m} (B_{ij} (V_i V_j \cos \theta_{ij} - V_i^{(0)} V_j^{(0)} \cos \theta_{ij}^{(0)})) \\ & + \sum_{i=1}^{n+m} \int G_{ij} V_i^2 d\theta_i + \sum_{i=1}^{n+m} \sum_{j=i+1}^{n+m} G_{ij} \int (V_i V_j \cos \theta_{ij} d(\theta_i + \theta_j) + \sin \theta_{ij} (V_j dV_i - V_i dV_j)) \\ & + \int \sum_{i=1}^{n+m} (P_{Li} d\theta_i) + \left(\frac{Q_{Li}}{V_i} \right) dV_i - \sum_{i=1}^n \int \left(P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i \right). \end{aligned}$$

It follows that

$$\begin{aligned} W_s + W_{PF} = & \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2 - \frac{1}{2} \sum_{i=1}^{n+m} B_{ii} (V_i^2 - V_i^{(0)2}) \\ & - \sum_{i=1}^{n+m-1} \sum_{j=i+1}^{n+m} B_{ij} (V_i V_j \cos \theta_{ij} - V_i^{(0)} V_j^{(0)} \cos \theta_{ij}^{(0)}) - \sum_{i=1}^n \int P_{mi} d\delta_i \\ & + \sum_{i=1}^n \int P_{ei} d\delta_i + \sum_{i=1}^n \int D_i \omega_i d\delta_i + W_{Gii} + W_{Gij} + W_{Load} \\ & - \sum_{i=1}^n \int \left(P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i \right). \end{aligned}$$

We define $W_{Gen} = \sum_{i=1}^n \int P_{ei} d\delta_i - \sum_{i=1}^n \int \left(P_{Gi} d\theta_i + \frac{Q_{Gi}}{V_i} dV_i \right)$, then by carrying out the integration terms, we have

$$\begin{aligned}
W_{gen} = & \frac{1}{4} \sum_{i=1}^n \frac{x''_{di} + x''_{qi}}{x''_{di} x''_{qi}} (V_i^2 - V_i^{0^2}) \\
& - \frac{1}{4} \sum_{i=1}^n \left(\frac{1}{x''_{qi}} - \frac{1}{x''_{di}} \right) \left(V_i^2 \cos 2(\delta_i - \theta_i) - V_i^{0^2} \cos 2(\delta_i^0 - \theta_i^0) \right) \\
& - \sum_{i=1}^n \int \left(\frac{x''_{di} - x_{ei}}{x''_{di} (x'_{di} - x_{ei})} e'_{gi} + \frac{(x_{di} - x'_{di})(x'_{di} - x''_{di}) + (x'_{di} - x_{ei})^2}{x''_{di} (x_{di} - x_{ei})(x'_{di} - x_{ei})} e''_{qi} \right) d(V_i \cos(\delta_i - \theta_i)) \\
& + \sum_{i=1}^n \int \left(\frac{x''_{qi} - x_{ei}}{x''_{qi} (x'_{qi} - x_{ei})} e'_{di} + \frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi}) + (x'_{qi} - x_{ei})^2}{x''_{qi} (x_{qi} - x_{ei})(x'_{qi} - x_{ei})} e''_{di} \right) d(V_i \sin(\delta_i - \theta_i)).
\end{aligned}$$

7.3.3 Energy Function Associated with Rotor Circuit Equations

An energy function associated with the rotor circuit equations can be obtained by the following procedure:

Step 1. Multiply both sides of $\frac{de'_{qi}}{dt}$ equation by $\frac{1}{x_{di} - x'_{di}} \left(\frac{de'_{gi}}{dt} \right)$.

Step 2. Multiply both sides of $\frac{de'_{gi}}{dt}$ equation by

$$\frac{(x_{di} - x'_{di})(x'_{di} - x''_{di}) + (x'_{di} - x_{ei})^2}{(x'_{di} - x''_{di})(x_{di} - x_{ei})^2} \left(\frac{de''_{qi}}{dt} \right).$$

Step 3. Multiply both sides of $\frac{de'_{gi}}{dt}$ equation by

$$\frac{(x_{di} - x'_{di})(x'_{di} - x''_{di}) + (x'_{di} - x_{ei})^2}{(x'_{di} - x''_{di})(x_{di} - x_{ei})^2} \left(\frac{de''_{qi}}{dt} \right).$$

Step 4. Multiply both sides of $\frac{de'_{di}}{dt}$ equation by $\frac{1}{x_{qi} - x'_{qi}} \left(\frac{de'_{di}}{dt} \right)$.

Step 5. Multiply both sides of $\frac{de'_{di}}{dt}$ equation by

$$\frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi}) + (x'_{qi} - x_{ei})^2}{(x'_{qi} - x''_{qi})(x_{qi} - x_{ei})^2} \left(\frac{de''_{di}}{dt} \right).$$

By summing over all the equations and integrating them, we get (after a tedious procedure) the energy function associated with generator and network equations, which can be summarized as follows:

$$W = W_s + W_{PF} + W_{Rotor}.$$

The above energy function can be rearranged as follows:

$$W = W_k + \sum_{i=1}^{23} U_i$$

W_k = kinetic energy

$$= \frac{1}{2} \sum_{i=1}^n M_i \omega_i^2$$

U_1 = potential energy due to network reactance

$$= -\frac{1}{2} \sum_{i=1}^{n+m} B_{ii} (V_i^2 - V_i^{02}) - \sum_{i=1}^{n+m-1} \sum_{j=i+1}^{n+m} B_{ij} (V_i V_j \cos \theta_{ij} - V_i^0 V_j^0 \cos \theta_{ij}^0)$$

U_2 = due to generator subtransient reactance

$$= \frac{1}{4} \sum_{i=1}^n \frac{x_{di}'' + x_{qi}''}{x_{di}'' x_{qi}''} (V_i^2 - V_i^{(0)2})$$

U_3 = due to $x_{di}'' \neq x_{qi}''$

$$= -\frac{1}{4} \sum_{i=1}^n \left(\frac{x_{di}'' - x_{qi}''}{x_{di}'' x_{qi}''} \right) \left(V_i^2 \cos 2(\delta_i - \theta_i) - V_i^{(0)2} \cos 2(\delta_i^{(0)} - \theta_i^{(0)}) \right)$$

U_4 = due to $e_{qi}' V_i$

$$= -\sum_{j=1}^n \int \frac{x_{di}'' - x_{ei}}{x_{di}'' (x_{di}' - x_{ei})} \left(e_{qi}' V_i \cos(\delta_i - \theta_i) - e_{qi}'^{(0)} V_i^{(0)} \cos(\delta_i^{(0)} - \theta_i^{(0)}) \right)$$

U_5 = due to $e_{qi}'' V_i$

$$= -\sum_{i=1}^n \int \frac{x_{di}'' - x_{ei}}{x_{di}'' (x_{di}' - x_{ei})} \left(1 + \frac{(x_{di} - x_{di}') (x_{di}' - x_{di}'')}{(x_{di}' - x_{ei})^2} \right) \left(e_{qi}'' V_i \cos(\delta_i - \theta_i) - e_{qi}''^{(0)} V_i^{(0)} \cos(\delta_i^{(0)} - \theta_i^{(0)}) \right)$$

U_6 = due to $e_{qi}'^2$

$$= \sum_{i=1}^n \frac{1}{2} \left(\frac{1}{x_{di} - x_{di}'} + \frac{x_{di}' - x_{di}''}{(x_{di}' - x_{ei})^2} + \frac{(x_{di}'' - x_{ei})^2}{x_{di}'' (x_{di}' - x_{ei})^2} \right) \left(e_{qi}'^2 - (e_{qi}'^{(0)})^2 \right)$$

U_7 = due to $e_{qi}''^2$

$$= \frac{1}{2} \sum_{i=1}^n \frac{x_{di}'}{x_{di}''} \frac{(x_{di}' - x_{ei})^2}{(x_{di}' - x_{di}'') (x_{di} - x_{ei})^2} \left(1 + \frac{(x_{di} - x_{di}') (x_{di}' - x_{di}'')}{(x_{di}' - x_{ei})^2} \right) \left(e_{qi}''^2 - (e_{qi}''^{(0)})^2 \right)$$

U_8 = due to $e_{qi}' e_{qi}''$

$$= -\sum_{i=1}^n \frac{x_{ei}}{e_{di}'' (x_{di} - x_{ei})} \left(1 + \frac{(x_{di} - x_{di}') (x_{di}' - x_{di}'')}{(x_{di}' - x_{ei})^2} \right) \left(e_{qi}' e_{qi}'' - e_{qi}'^{(0)} e_{qi}''^{(0)} \right)$$

U_9 = due to $e_{di}' V_i$

$$= \sum_{i=1}^n \frac{x_{qi}'' - x_{ei}}{x_{qi}'' (x_{qi}' - x_{ei})} \left(e_{di}' V_i \sin(\delta_i - \theta_i) - e_{di}'^{(0)} V_i^{(0)} \sin(\delta_i^{(0)} - \theta_i^{(0)}) \right)$$

$$\begin{aligned}
 U_{10} &= \text{due to } e''_{di} V_i \\
 &= \sum_{i=1}^n \frac{(x'_{qi} - x_{ei})}{x''_{qi}(x_{qi} - x_{ei})} \left(1 + \frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi})}{(x'_{qi} - x_{ei})^2} \right) \\
 &\quad (e''_{di} V_i \sin(\delta_i - \theta_i) - e''_{di(0)} V_i^{(0)} \sin(\delta_i^{(0)} - \theta_i^{(0)}))
 \end{aligned}$$

$$\begin{aligned}
 U_{11} &= \text{due to } e'^2_{di} \\
 &= \sum_{i=1}^n \frac{1}{2} \left(\frac{1}{x_{qi} - x'_{qi}} + \frac{x'_{qi} - x''_{qi}}{(x'_{qi} - x_{ei})^2} + \frac{(x''_{qi} - x_{ei})^2}{x''_{qi}(x'_{qi} - x_{ei})^2} \right) (e'^2_{di} - (e'^{(0)}_{di})^2)
 \end{aligned}$$

$$\begin{aligned}
 U_{12} &= \text{due to } e''^2_{di} \\
 &= \frac{1}{2} \sum_{i=1}^n \frac{x'_{qi}}{x''_{qi}} \frac{(x'_{qi} - x_{ei})^2}{(x'_{qi} - x''_{qi})(x_{qi} - x_{ei})^2} \left(1 + \frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi})}{(x'_{qi} - x_{ei})^2} \right)^2 (e''^2_{di} - (e''^{(0)}_{di})^2)
 \end{aligned}$$

$$\begin{aligned}
 U_{13} &= \text{due to } e'_{di} e''_{di} \\
 &= - \sum_{i=1}^n \frac{x_{ei}}{x''_{qi}(x_{qi} - x_{ei})} \left(1 + \frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi})}{(x'_{qi} - x_{ei})^2} \right) (e'_{di} e''_{di} - e'^{(0)}_{di} e''^{(0)}_{di})
 \end{aligned}$$

$$\begin{aligned}
 U_{14} &= \text{due to conductance} \\
 &= \sum_{i=1}^{n+m} G_{ii} \int V_i^2 d\theta_i + \sum_{i=1}^{n+m-1} \sum_{j=i+1}^{n+m} G_{ij} \int (V_i V_j \cos \theta_{ij} d(\theta_i + \theta_j) + \sin \theta_{ij} (V_j dV_i - V_i dV_j))
 \end{aligned}$$

$$U_{15} = \text{due to active load} = \sum_{i=1}^{n+m} \int (P_{Li} d\theta_i)$$

$$U_{16} = \text{due to active load} = \sum_{i=1}^{n+m} \int \frac{Q_{Li}}{V_i} dV_i$$

$$U_{17} = \text{due to mechanical power} = - \sum_{i=1}^n \int P_{mi} d\delta_i$$

$$U_{18} = \text{due to damping} = \sum_{i=1}^n D_i \int \omega_i d\delta_i$$

$$U_{19} = \text{due to excitation} = - \sum_{i=1}^n \frac{1}{x_{di} - x'_{di}} \int E_{fdi} dE'_{gi}$$

$$U_{20} = \text{due to } \frac{de'_{qi}}{dt} \text{ variation} = \sum_{i=1}^n \frac{T'_{doi}}{x_{di} - x'_{di}} \int \left(\frac{de'_{gi}}{dt} \right)^2 dt$$

$$\begin{aligned}
 U_{21} &= \text{due to } \frac{de''_{qi}}{dt} \\
 &= \sum_{i=1}^n \frac{T''_{dki}}{x'_{di} - x''_{di}} \frac{(x_{di} - x'_{di})(x'_{di} - x''_{di}) + (x'_{di} - x_{ei})^2}{(x_{di} - x_{ei})^2} \left(\frac{de''_{qi}}{dt} \right)^2 dt
 \end{aligned}$$

$$U_{22} = \text{due to } \frac{de'_{di}}{dt} = \sum_{i=1}^n \frac{T'_{qoi}}{x_{qi} - x'_{qi}} \int \left(\frac{de'_{di}}{dt} \right)^2 dt$$

$$U_{23} = \text{due to } \frac{de''_{di}}{dt} = \sum_{i=1}^n \frac{T''_{qoi}}{x'_{qi} - x''_{qi}} \frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi}) + (x'_{qi} - x_{ei})^2}{(x_{qi} - x_{ei})^2} \left(\frac{de''_{di}}{dt} \right)^2 dt.$$

Note that $U_1 - U_{13}$ are path-independent terms that have analytical forms; $U_{14} - U_{23}$ are path-dependent terms for which numerical integration is required by proper path assumptions.

Because the derivation of the above energy function is based on the first integral, therefore, we have $\frac{dW}{dt} = 0$ along the postfault trajectory. If the terms $U_{18} \cdot U_{20} \cdot U_{21} \cdot U_{22} \cdot U_{23}$ are not included in the energy function, then along the postfault trajectory, we have

$$\begin{aligned} \frac{dW}{dt} = & -\sum_{i=1}^n D_i \omega_i^2 - \sum_{i=1}^n \frac{T'_{doi}}{x_{di} - x'_{di}} \int \left(\frac{de'_{gi}}{dt} \right)^2 - \sum_{i=1}^n \frac{T'_{qoi}}{x_{qi} - x'_{qi}} \int \left(\frac{de'_{gi}}{dt} \right)^2 \\ & - \sum_{i=1}^n \frac{T''_{dki}}{x'_{di} - x''_{di}} \frac{(x_{di} - x'_{di})(x'_{di} - x''_{di}) + (x'_{di} - x_{ei})^2}{(x_{di} - x_{ei})^2} \left(\frac{de''_{gi}}{dt} \right)^2 \\ & - \sum_{i=1}^n \frac{T''_{qki}}{x'_{qi} - x''_{qi}} \frac{(x_{qi} - x'_{qi})(x'_{qi} - x''_{qi}) + (x'_{qi} - x_{ei})^2}{(x_{qi} - x_{ei})^2} \left(\frac{de''_{di}}{dt} \right)^2 \leq 0. \end{aligned}$$

7.4 ILL-CONDITIONED NUMERICAL PROBLEMS

A numerical ill-conditioning problem can occur in the numerical energy function by the ray approximation scheme. By observing Equations 7.7 and 7.8, it is clear that an ill-conditioning problem will occur if $\Delta\theta_{ij}$ is close to zero. We illustrate this ill-conditioning problem on the following nine-bus system, shown in Figure 7.3. We consider the case of nonuniform damping: $Di/Mi = [0.1, 0.2, 0.3]$ with the constant impedance load (see Table 7.1).

We summarize the postfault stable equilibrium point (SEP) and controlling unstable equilibrium point (UEP) for Contingency 1 in Table 7.2. The angle differences between the controlling UEP and the postfault SEP are needed for evaluating the energy margin in the controlling UEP method. The angle differences between the exit point and the postfault SEP needed for evaluating the energy margin in the potential energy boundary surface (PEBS) method are listed in Table 7.3. It is noticed that the angle difference, $\Delta\theta_{45} = \Delta\theta_4 - \Delta\theta_5$, is almost zero, which will cause a numerical ill-conditioning problem in the computation of potential energy related to transfer conductance based on the ray approximation scheme.

We summarize the postfault SEP and controlling UEP for Contingency 2 in Table 7.4. The angle differences between the controlling UEP and the postfault SEP are needed for evaluating the energy margin in the controlling UEP method. The angle differences between the exit point and the postfault SEP needed for evaluating

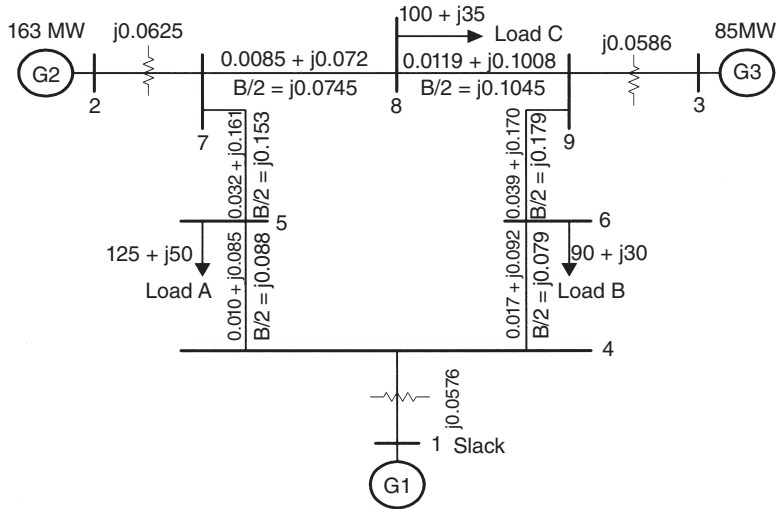


Figure 7.3 A three-generator, nine-bus test system.

Table 7.1 Contingency List and Postfault SEPs

Contingency number	Faulted bus	Fault clearing type	Description from bus to bus	Postfault SEPs ($\delta_1, \delta_2, \delta_3$)
1	7	Line trip	7 5	(-0.1736, 0.5199, 0.2576)
2	7	Line trip	8 7	(-0.0987, 0.3980, -0.0708)
4	5	Line trip	5 4	(-0.0590, 0.1786, 0.0839)
5	4	Line trip	4 6	(-0.0592, 0.1957, 0.0492)
7	9	Line trip	6 9	(-0.1322, 0.3365, 0.3230)
8	9	Line trip	9 8	(-0.0764, 0.2031, 0.1682)

Table 7.2 Postfault SEP and Controlling UEP for Contingency 1

	($\delta_1, \delta_2, \delta_3$)	($V_1, V_2, \dots, V_9$)	($\theta_1, \theta_2, \dots, \theta_9$)
x_{sep}	(-0.1736, 0.5199, 0.2576)	(1.0141, 1.0067, 0.9955, 0.9748, 0.9257, 0.9544, 0.9973, 0.9796, 0.9930)	(-0.2065, 0.3426, 0.1142, -0.2404, -0.3383, -0.1834, 0.2456, 0.1319, 0.0669)
x_{cuep}	(-0.7657, 2.0560, 1.6419)	(0.8276, 0.8436, 0.6418, 0.6106, 0.5799, 0.3227, 0.7490, 0.6381, 0.5272)	(-0.7641, 1.8889, 1.4835, -0.7616, -0.8596, -0.3516, 1.7669, 1.6004, 1.3850)

the energy margin in the PEBS method are listed in Table 7.5. We notice that the angle difference, $\Delta\theta_{89} = \Delta\theta_8 - \Delta\theta_9$, is almost zero, which will cause a numerical ill-conditioning problem in the computation of potential energy related to transfer conductance based on the ray approximation scheme.

Table 7.3 Angle Differences for Contingency 1

	$\Delta\delta_1$	$\Delta\delta_2$	$\Delta\delta_3$	$\Delta\theta_1$	$\Delta\theta_2$	$\Delta\theta_3$	$\Delta\theta_4$	$\Delta\theta_5$	$\Delta\theta_6$	$\Delta\theta_7$	$\Delta\theta_8$	$\Delta\theta_9$
$\theta_{cusp} - \theta_{sep}$	-0.5921	1.5360	1.3843	-0.557	1.5462	1.3693	-0.5212	-0.5212	-0.1682	1.5213	1.4685	1.3181
$\theta_{exit} - \theta_{sep}$	-0.5404	1.8362	0.3400	-0.508	1.7496	0.4448	-0.4763	-0.4563	-0.3134	1.5708	1.1733	0.4645

The angle difference, $\Delta\theta_{4,5} = \Delta\theta_4 - \Delta\theta_5$, is almost zero, which will cause a numerical ill-conditioning problem.

Table 7.4 Postfault System SEP and Controlling UEP for Contingency 2

	$[\delta_1, \delta_2, \delta_3]$	$[V_1, V_2, \dots, V_9]$	$[\theta_1, \theta_2, \dots, \theta_9]$
x_{sep}	$[-0.0987, 0.3980, -0.0708]$	$[1.0289, 1.0145, 0.9893, 1.0038, 0.9647, 0.9835, 1.0092, 0.9476, 0.9851]$	$[-0.1331, 0.2200, -0.2174, -0.1674, -0.1342, -0.2530, 0.1237, -0.3576, -0.2662]$
x_{cuep}	$[-0.4879, 2.0301, -0.4842]$	$[0.8697, 0.6270, 0.8816, 0.6926, 0.3936, 0.7334, 0.4405, 0.8093, 0.8413]$	$[-0.4934, 1.7938, -0.6074, -0.5013, -0.2621, -0.6115, 1.4984, -0.7469, -0.6555]$

We summarize the postfault SEP and the controlling UEP for Contingency 4 in Table 7.6. The angle differences between the controlling UEP and the postfault SEP are needed for evaluating the energy margin in the controlling UEP method. The angle differences between the exit point and the postfault SEP needed for evaluating the energy margin in the PEBS method are listed in Table 7.7. We note that the angle difference, $\Delta\theta_{57} = \Delta\theta_5 - \Delta\theta_7$, is almost zero, which will cause a numerical ill-conditioning problem in the computation of potential energy related to transfer conductance based on the ray approximation scheme.

The numerical ill-conditioning problem using the ray approximation scheme can be eliminated by using the following rule:

$$\text{If } \Delta\theta_{ij} = \theta_{ij} - \theta_{ij0} < 1.0e^{-6}, \text{ then assume } \Delta\theta_{ij} = 0.0.$$

Then, the path-dependent terms U_2 and U_3 can be rewritten by the modified ray approximation in the case that $\Delta\theta_{ij} \approx 0.0$:

$$U_2 \approx \sum_{i=1}^{n-1} \sum_{j=i+1}^n G_{ij} (\Delta\theta_i + \Delta\theta_j) \cos \theta_{ij0} \left[V_{i0} V_{j0} + \frac{1}{2} (V_{i0} \Delta V_j + V_{j0} \Delta V_i) + \frac{1}{3} \Delta V_i \Delta V_j \right] \quad (7.14)$$

$$U_3 \approx \sum_{i=1}^{n-1} \sum_{j=i+1}^n G_{ij} \Delta V_i \sin \theta_{ij0} \left(\frac{V_j + V_{j0}}{2} \right). \quad (7.15)$$

7.5 NUMERICAL EVALUATIONS OF APPROXIMATION SCHEMES

To evaluate the numerical energy function using the ray approximation scheme and that using the trapezoidal approximation scheme, we apply the controlling UEP method to these two numerical energy functions to direct the estimation of CCTs. The estimated CCTs are then compared to the exact CCTs based on the time-domain simulation. A summary of the comparison between the exact CCTs and the estimated CCTs is presented in Tables 7.8 and 7.9.

Table 7.8 lists for each contingency the potential energy at the exit point used as the critical energy by the PEBS method, the potential energy at the controlling UEP used as the critical energy by the controlling UEP method, the CCT based on the controlling UEP method with the numerical energy function evaluated via the

Table 7.5 Angle Differences for Contingency 2

	$\Delta\delta_1$	$\Delta\delta_2$	$\Delta\delta_3$	$\Delta\theta_1$	$\Delta\theta_2$	$\Delta\theta_3$	$\Delta\theta_4$	$\Delta\theta_5$	$\Delta\theta_6$	$\Delta\theta_7$	$\Delta\theta_8$	$\Delta\theta_9$
$\theta_{cuep} - \theta_{sep}$	-0.3892	1.6321	-0.4134	-0.3603	1.5738	-0.3900	-0.3339	-0.1279	-0.3585	1.3747	-0.3892	-0.3892
$\theta_{exit} - \theta_{sep}$	-0.5790	1.8280	0.6604	-0.4863	1.8447	0.4377	-0.3529	-0.1629	-0.0919	1.7284	0.3203	0.3203

$\Delta\theta_{89} = \Delta\theta_8 - \Delta\theta_9$ is almost zero, which will cause a numerical ill-conditioning problem in the computation of potential energy related to transfer conductance based on the ray approximation scheme.

Table 7.6 Postfault System SEP, Point, and Controlling UEP (CUEP) for Contingency 4

	$[\delta_1, \delta_2, \delta_3]$	$[V_1, V_2, \dots, V_9]$	$[\theta_1, \theta_2, \dots, \theta_9]$
SEP	$[-0.0590, 0.1786, 0.0839]$	$[1.0499, 0.9950, 1.0146, 1.0444, 0.8798, 1.0208, 0.9797, 0.9836, 1.0180]$	$[-0.0895, 0.0008, -0.0551, -0.1187, -0.2714, -0.1638, -0.0983, -0.1384, -0.0999]$
CUEP	$[-0.8563, 2.2192, 2.0069]$	$[0.8265, 0.8137, 0.5975, 0.6084, 0.6333, 0.2638, 0.7052, 0.5974, 0.4703]$	$[-0.8564, 2.0440, 1.8339, -0.8566, 1.7351, -0.6079, 1.9082, 1.8060, 1.7133]$

modified ray approximation, the exact CCT based on the time-domain simulation, and the associated error. As observed from the error information provided in this table, the large errors suggest that the ray approximation scheme is not acceptable for constructing numerical energy functions.

Table 7.9 lists for each contingency the potential energy at the exit point used as the critical energy by the PEBS method, the potential energy at the controlling UEP used as the critical energy by the controlling UEP method, the CCT based on the controlling UEP method with the numerical energy function evaluated via the trapezoidal approximation scheme, the exact CCT based on the time-domain simulation, and the associated error. As observed from the error information provided in this table, the small errors suggest that the trapezoidal approximation scheme is acceptable for constructing numerical energy functions from which the trapezoidal approximation scheme seems acceptable in this numerical study.

Even though the modified ray approximation scheme can eliminate the numerical ill-conditioning problem, the estimated CCTs obtained using the modified ray approximation are significantly different from the exact CCTs. In addition, it is observed that in most contingencies of the test system, the numerical energy function based on the modified ray approximation scheme leads to overestimates in CCTs. On the other hand, the trapezoidal approximation scheme does not suffer from the numerical ill-conditioned problem, and this scheme leads to conservative estimates of CCTs for all contingency cases in this numerical study, which shows that the trapezoidal approximation scheme outperforms the ray approximation scheme in approximating path-dependent terms. To further improve the trapezoidal approximation scheme, an improved trapezoidal approximation scheme will be presented in the next section.

7.6 MULTISTEP TRAPEZOIDAL SCHEME

An improved approximation scheme for evaluating numerical energy functions is presented in this section. The basic idea of this improved scheme is the following: instead of evaluating the energy function value at the controlling UEP by applying the trapezoidal approximation scheme to the two state vectors, the controlling UEP and the postfault SEP, the evaluation is performed by applying the trapezoidal rule

Table 7.7 Angle Differences for Contingency 4

	$\Delta\delta_1$	$\Delta\delta_2$	$\Delta\delta_3$	$\Delta\theta_1$	$\Delta\theta_2$	$\Delta\theta_3$	$\Delta\theta_4$	$\Delta\theta_5$	$\Delta\theta_6$	$\Delta\theta_7$	$\Delta\theta_8$	$\Delta\theta_9$
$\theta_{curep} - \theta_{sep}$	-0.7973	2.0406	1.9230	-0.7670	2.0433	1.8890	-0.7380	2.0065	-0.4441	2.0065	1.9444	1.8132
$\theta_{exit} - \theta_{sep}$	-0.7919	2.1944	1.5537	-0.7435	2.1346	1.5821	-0.6859	2.0377	-0.3282	2.0377	1.8762	1.5487

$\Delta\theta_{s7} = \Delta\theta_s - \Delta\theta_7$ is almost zero, which will cause a numerical ill-conditioning problem in the computation of potential energy related to transfer conductance based on the ray approximation scheme.

Table 7.8 CCT Estimates Using the Numerical Energy Function Evaluated Via the Modified Ray Approximation

Contingency number	Potential energy at exit point	Potential energy at CUEP	CCT estimate	Exact CCT	Error (%)
1	1.8234	3.5802	0.262	0.170	-54.12
2	1.8975	-0.2992	0.0000	0.187	100.0
4	9.2757	9.5050	0.520	0.424	-22.64
5	7.8330	7.6405	1.000	0.320	-212.5
7	0.9867	2.1678	0.259	0.224	-15.63
8	2.6840	1.0988	0.174	0.245	28.98

The large errors suggest that the ray approximation scheme is not acceptable for constructing numerical energy functions.

Table 7.9 CCT Estimates Using the Numerical Energy Function Evaluated Via the Trapezoidal Approximation Scheme

Contingency number	Potential energy at exit point	Potential energy at CUEP	CCT estimate	Exact CCT	Error (%)
1	1.0625	0.7663	0.158	0.170	7.06
2	1.7909	0.9465	0.155	0.187	17.11
4	2.3545	2.2614	0.411	0.424	3.07
5	2.4229	2.1744	0.304	0.320	5.00
7	1.9412	1.0476	0.194	0.224	13.40
8	2.6094	1.7112	0.207	0.245	15.51

The small errors suggest that the trapezoidal approximation scheme is acceptable for constructing numerical energy functions.

to multiple points between the controlling UEP and the postfault SEP (see Figure 7.4).

For instance, the three-step trapezoidal approximation scheme applies the trapezoidal rule to the controlling UEP, x_1 , x_2 , and the postfault SEP as the “end points” to evaluate the path-dependent integration via the following three numerical integrations:

$$\int_{x_s^{post}}^{x_{co}} f(x) dx \cong \frac{b-a_1}{2}(f(a_1)+f(b)) + \frac{a_1-a_2}{2}(f(a_1)+f(a_2)) + \frac{a_2-a}{2}(f(a_2)+f(a)) \quad (7.16)$$

where $b = f(x_{co})$, $a_1 = f(x_1)$, $a_2 = f(x_2)$, $a = f(x_s^{post})$.

This multistep trapezoidal approximation scheme provides a more accurate numerical energy function evaluation. When applied to the nine-bus system with six contingencies, it also leads to more accurate CCT estimates for each contingency.

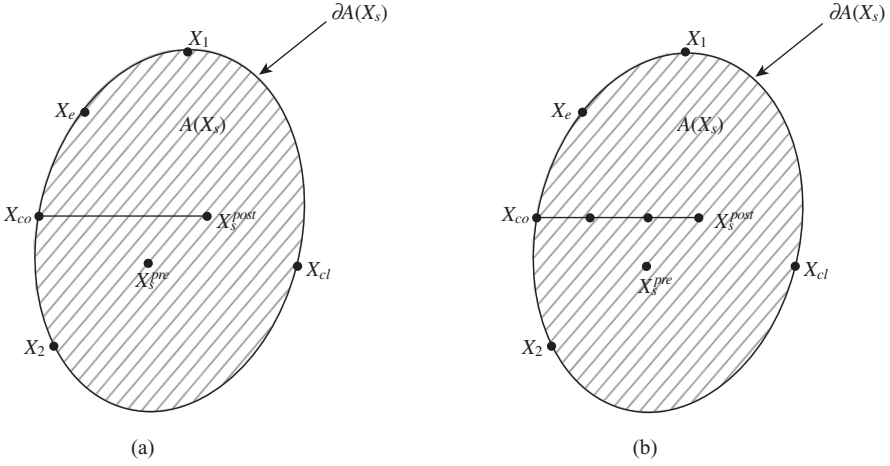


Figure 7.4 (a) The traditional trapezoidal approximation scheme applies the trapezoidal rule to the controlling UEP and the postfault SEP as the “end points” to evaluate the path-dependent integration. (b) The three-step trapezoidal approximation scheme applies the trapezoidal rule to the controlling UEP, x_1 , x_2 , and the postfault SEP as the “end points” to evaluate the path-dependent integration via three numerical integrations.

A summary of the comparison among the exact CCTs using the time-domain simulation, the estimated CCTs using the trapezoidal approximation scheme, and the (new) estimated CCTs using the multistep trapezoidal approximation scheme are presented in Table 7.9. The errors of estimated CCTs using the multistep trapezoidal approximation scheme, as compared with the exact CCTs shown in the table, are considerably smaller than the errors using the traditional trapezoidal approximation scheme. For example, the error for Contingency 1 is reduced from 7.06% to 1.18%, and from 3.07% to 0.24% for Contingency 4. The error for Contingency 5 is reduced from 5.00% to 1.88% and from 15.51% to 9.8% for Contingency 8. Hence, the multistep trapezoidal approximation scheme leads to accurate estimated CCTs as compared with the traditional trapezoidal approximation scheme for each contingency.

The number of steps used in the multistep trapezoidal approximation scheme impacts on the energy function value at the point of evaluation such as at the controlling UEP. As the number of steps increases, the energy function value at the point of evaluation also improves at the expense of minor computational efforts. As the number of steps increases to a threshold value, the energy function value at the point of evaluation saturates. As listed in Table 7.10, the CCT estimate is based on the traditional one, while the new CCT estimate is based on the multistep one. The error is associated with the multistep trapezoidal approximation scheme. For instance, the two-step trapezoidal approximation scheme yields the numerical energy function value of 0.8385 at the controlling UEP, while the corresponding CCT estimate is 0.165 s.

Table 7.10 CCT Estimates Using the Numerical Energy Function Evaluation Via the Multistep Trapezoidal Approximation Scheme

Contingency number	Potential energy at exit point	Improved potential energy at CUEP	CCT estimate	New CCT estimate	Exact CCT	Error (%)
1	1.0625	0.8733	0.158	0.168	0.170	1.18
2	1.7909	1.1251	0.155	0.167	0.187	10.7
4	2.3545	2.3934	0.411	0.425	0.424	0.24
5	2.4229	2.3528	0.304	0.314	0.320	1.88
7	1.9412	1.1908	0.194	0.204	0.224	8.93
8	2.6094	2.0343	0.207	0.221	0.245	9.80

Table 7.11 The Number of Point Evaluation, the Variation of Potential Energy Related to Path-Dependent Terms, the Variation of Potential Energy at CUEP, and CCT Estimates for Contingency 1

Number of integration steps	Path-dependent potential energy	Total potential energy	CCT estimate (s)
1	0.7238	0.7663	0.158
2	0.796	0.8385	0.165
3	0.8149	0.8574	0.166
5	0.8251	0.8676	0.167
7	0.8279	0.8704	0.168
10	0.8294	0.8719	0.168
15	0.8302	0.8727	0.168
20	0.8305	0.873	0.168
30	0.8307	0.8732	0.168
50	0.8308	0.8733	0.168
80	0.8308	0.8733	0.168
100	0.8308	0.8733	0.168

The improvement in estimated CCTs and in critical energy evaluation increases with the number of steps in the evaluation.

We next perform numerical studies for evaluating the impacts of numbers of steps in the multistep trapezoidal approximation scheme on CCT estimations. The number of steps starts from 1, and then 2, and then 3, 5, 7, 10, 15, 20, 30, 50, 80, and finally 100. The corresponding estimated CCTs are listed in Tables 7.11–7.14 for four different contingencies, respectively. As shown in Table 7.11, the traditional trapezoidal approximation scheme (i.e., the one-step scheme) yields the numerical energy function value of 0.7663 at the controlling UEP, and the corresponding CCT estimate is 0.158 s. The two-step trapezoidal approximation scheme yields the

Table 7.12 The Number of Point Valuation, the Variation of Potential Energy Related to Path-Dependent Terms, the Variation of Potential Energy at CUEP, and CCT Estimates for Contingency 2

Number of integration steps	Path-dependent potential energy	Total potential energy	CCT estimate (s)
1	-0.3974	0.9465	0.155
2	-0.2752	1.0686	0.163
3	-0.2446	1.0993	0.165
5	-0.2281	1.1157	0.166
7	-0.2235	1.1203	0.167
10	-0.221	1.1228	0.167
15	-0.2197	1.1241	0.167
20	-0.2193	1.1246	0.167
30	-0.2189	1.1249	0.167
50	-0.2188	1.1251	0.167
80	-0.2187	1.1251	0.167
100	-0.2187	1.1251	0.167

The improvement in estimated CCTs and in critical energy evaluation increases with the number of steps in the evaluation, but this improvement saturates when the number of steps equals seven and beyond.

Table 7.13 The Number of Point Evaluation, the Variation of Potential Energy Related to Path-Dependent Terms, the Variation of Potential Energy at CUEP, and CCT Estimates for Contingency 4

Number of integration steps	Path-dependent potential energy	Total potential energy	CCT estimate (s)
1	2.7657	2.2614	0.411
2	2.8428	2.3385	0.419
3	2.8731	2.3687	0.422
5	2.8893	2.385	0.424
7	2.8936	2.3892	0.424
10	2.8958	2.3914	0.425
15	2.8969	2.3926	0.425
20	2.8973	2.3929	0.425
30	2.8976	2.3932	0.425
50	2.8977	2.3934	0.425
80	2.8978	2.3934	0.425
100	2.8978	2.3934	0.425

The improvement in estimated CCTs and in critical energy evaluation increases with the number of steps in the evaluation, but this improvement saturates when the number of steps equals five and beyond.

Table 7.14 The Number of Point Evaluation, the Variation of Potential Energy Related to Path-Dependent Terms, the Variation of Potential Energy at CUEP, and CCT Estimates for Contingency 5

Number of integration steps	Path-dependent potential energy	Total potential energy	CCT estimate (s)
1	2.3019	2.1744	0.304
2	2.4075	2.2799	0.31
3	2.4479	2.3204	0.312
5	2.4693	2.3418	0.313
7	2.4749	2.3473	0.314
10	2.4777	2.3502	0.314
15	2.4792	2.3517	0.314
20	2.4797	2.3522	0.314
30	2.4801	2.3526	0.314
50	2.4803	2.3528	0.314
80	2.4803	2.3528	0.314
100	2.4803	2.3528	0.314

The improvement in estimated CCTs and in critical energy evaluation increases with the number of steps in the evaluation, but this improvement saturates when the number of steps equals seven and beyond.

numerical energy function value of 0.8385 at the controlling UEP and the corresponding CCT estimate is 0.165 s. The three-step trapezoidal approximation scheme yields the numerical energy function value of 0.8574 at the controlling UEP and the corresponding CCT estimate is 0.166 s. The improvement in the estimated CCTs and in the critical energy values increases with the number of steps in the evaluation, but this improvement saturates when the number of steps equals seven and beyond.

Similar improvements also occur with other contingencies as summarized in Tables 7.12–7.14, respectively. The claim that the improvement in the estimated CCTs and in the critical energy evaluation increases with the number of steps in the evaluation but that this improvement saturates when the number of steps reaches a threshold still holds. These threshold values vary among different contingencies. For example, the threshold value is 7 for Contingencies 1 and 2, 5 for Contingency 4, and 7 for Contingency 5. The determination of this threshold value seems contingency dependent.

7.7 ON THE CORRECTED NUMERICAL ENERGY FUNCTIONS

Since there may not exist an analytical energy function for lossy transient stability models, the issue of verifying numerical energy functions and of correcting these numerical energy functions when they fail to provide accurate energy margins arises.

This issue has not received serious treatment. In the literature, one attempt to correct energy margin by using physical arguments to adjust the kinetic energy of those generators that contribute to the system separation was proposed. The accuracy of this heuristic method is still unclear. Because there are presently no better alternatives, the numerical energy functions constructed via explicit trajectory approximations are still in widespread use.

There are some other numerical difficulties brought about by these path-dependent terms in the potential energy function. From a theoretical viewpoint, energy function values should be evaluated with respect to a reference point—the postfault SEP. However, the simulated trajectory between the postfault SEP and the point in the state space whose energy value is to be evaluated requires a full time-domain simulation, which is time-consuming. One alternative is to compute and store energy values along the fault-on trajectory. Once the critical energy is determined, which is the energy value at the controlling UEP using the linear path from prefault SEP to the controlling UEP, the estimated CCT can be obtained by comparing the computed energy margin to the stored energy values along the fault-on trajectory. However, the accuracy of this CCT computing procedure can be severely damaged by the path-dependent terms in the energy function. This is because the critical energy margin is computed by using the linear path trajectory (i.e., the ray approximation) from the prefault SEP to the controlling UEP, while the energy values along the fault-on trajectory are computed using the actual fault-on trajectory. The problem is that the fault-on trajectory may be way off the linear path from the prefault SEP and the fault clearing point, making the error of the resulting numerical energy function unacceptable. Apparently, some correction to the computed energy margin must be made to the goal that the corrected energy margin is accurate and conservative.

It is believed that a corrective scheme to be added into the current procedure of computing energy margins can be developed. The purpose of the corrective scheme is to compensate the effect of the assumed linear path or the quadratic path in computing the energy margin. One approach to the development of the corrective scheme is the usage of the actual fault-on trajectory instead of the linear path. This approach deserves further investigation.

7.8 CONCLUDING REMARKS

Two general methods for deriving numerical energy functions have been presented in this chapter. One method is based on the so-called first integral principle, while the other is a two-step procedure. The existing numerical energy functions all contain two parts: the analytical terms (i.e., path-independent terms) and the path-dependent terms. The path-dependent terms are computed based on either the so-called ray approximation scheme or the trapezoidal approximation scheme. These two approximation schemes have been examined.

In this chapter, it has been shown that a numerical ill-conditioned problem may arise in the evaluation of path-dependent potential energy using the ray

approximation scheme. A modified ray approximation scheme has been proposed to overcome the problem. However, the estimated CCTs obtained using this modified approximation are still significantly different from the exact CCTs. It is observed from numerical studies that the modified ray approximation scheme can lead to overestimates in CCTs. On the other hand, the trapezoidal approximation scheme does not suffer from the ill-conditioned problem, and this scheme leads to conservative estimates of CCTs for all contingency cases in this numerical study. Hence, the trapezoidal approximation scheme seems to outperform the ray approximation scheme in the evaluation of path-dependent potential energy.

To further improve the trapezoidal approximation scheme, a multistep trapezoidal approximation scheme has been presented. One significant feature of the improved scheme is that the error of the estimated CCTs using the multistep trapezoidal approximation scheme, when compared to the exact CCTs, is considerably smaller than the error from the traditional trapezoidal approximation scheme. The improvement in the estimated CCTs and in the critical energy evaluation increases with the number of steps in the multistep trapezoidal approximation scheme, but this improvement plateaus when the number of steps reaches a threshold value. The determination of this threshold value is contingency dependent.

Since there are presently no better alternatives, the numerical energy functions constructed via explicit trajectory approximations are still in widespread use. To correct the possible error brought about by numerical energy functions, corrections to the computed energy functions must be made such that the corrected energy function leads to accurate and conservative stability assessment results. The idea of adding a corrective scheme to the current procedure for computing energy margins seems promising. The purpose of this corrective scheme is to compensate the effect of an assumed linear or quadratic path when computing energy functions. One effective approach to the development of this corrective scheme is the usage of the actual fault-on trajectory instead of the linear path since the actual fault-on trajectory is available in direct methods. Nevertheless, this approach deserves further investigation.